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ORIGINAL INVESTIGATION

Effects of Position and Injury Status on Associations Between Preseason Workload and Heart-Rate Variability Profiles in American College Football Players

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Purpose: To quantify position-specific associations between heart-rate-variability profile characteristics and external-workload parameters in American college football players during preseason camp while accounting for injury-related participation status. ***Methods:*** Twenty-seven players categorized into skill (n = 15) and midskill (n = 12) position groups participated in the study. Players were further categorized as having "full go" (n = 15) or "go as can" (n = 12) participation status. The camp mean and coefficient of variation (CV) (indexing daily variation) for various workload parameters and resting heart rate and the natural log of the root mean square of successive differences (LnRMSSD) were quantified. ***Results:*** Skill players who consistently performed higher total workloads (total distance and player load) exhibited the most stable LnRMSSD (ie, lower CV; $P < .05$), while midskill players with the highest total workloads exhibited greater daily variation in LnRMSSD (ie, higher CV; $P < .05$). These associations persisted after injury-status adjustment. However, consistent between position groups, players with greater daily variation in select workload parameters also showed greater daily variation in LnRMSSD ($P < .05$). No between-groups differences for injury status were observed ($P > .05$), although effect-size analysis showed that LnRMSSD was *moderately* greater in "full go" players (effect size = 0.73, $P < .07$). ***Conclusions:*** Daily LnRMSSD variation better reflected position-specific workload responses, whereas mean LnRMSSD may affect or be affected by injury status. While skill players tolerated high total workloads with stable LnRMSSD, midskill players exhibited less day-to-day variation in LnRMSSD and high-intensity outputs at moderate workloads.

Keywords: autonomic, cardiovascular, heat, sport science, sports medicine

Preseason collegiate American football camp exposes players to an annual spike in training load due to lengthier training sessions, twice-a-day practices, and limited passive recovery days.[1] In addition to wear-and-tear on the body from intense training demands, physiological stress is enhanced by hot and humid weather conditions, protective equipment requirements, and for some players, high body mass and inadequate aerobic fitness.[2,3] Although high training loads are intended to improve physical capacities, they come with an increased risk of overreaching and injury.[4] Heterogeneity in physical (eg, body mass and composition) and performance (eg, aerobic fitness) characteristics among players and position groups[5] may further contribute to interindividual variation in training responses.[6] Thus, an important area of investigation concerns practical monitoring strategies that assist performance staff in identifying and mitigating negative responses to training.

Wearable movement trackers objectively quantify external workloads via global positioning systems and tri-axial accelerometers. Implementation of this technology has become ubiquitous among college football teams to monitor mechanical and locomotor demands of practice and competition.[1] These devices generate a diverse output of workload parameters representative of the training dose while physiological indicators of adaptation represent training responses. Together, these elements enable the determination of dose–response relationships. Regarding physiological indicators, resting heart rate (RHR) and vagal-mediated heart-rate variability (HRV) are accessible markers of cardiac-autonomic status that are sensitive to acute and chronic effects of competitive football, including some that may be medically relevant.[7] Established HRV profile characteristics include daily measures averaged over a period of time (eg, 1 wk, a camp) and the coefficient of variation (CV), which reflects the magnitude of daily variation.[8] These parameters better reflect autonomic status than isolated values[9] but have yet to be investigated during preseason football camp.

Higher HRV averages and lower CV values typically reflect positive training responses such as improvements in cardiorespiratory fitness. For example, rugby players with the most stable HRV amid peak workloads preceding the Olympics demonstrated the greatest improvements in maximum aerobic speed.[8] Nevertheless, only one previous study investigated HRV profile characteristics in a college football team. Results showed that players who performed the greatest total workloads during spring camp had the lowest CV values, consistent with the notion that a capacity for higher workloads protects against fluctuations in HRV.[6] However, noteworthy limitations were as follows: position-specific associations were not quantified; weather conditions were more temperate relative to summer; training sessions were less frequent than preseason camp; and movement parameter assessment was limited to a single metric of total workload.[6] These shortcomings of prior work highlight a need for further study during preseason camp.

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Position group may affect associations between external workloads and HRV due to differences in playing demands, body mass, and aerobic fitness.[1,5] For instance, the skill group includes receivers and defensive backs who tend to be physically smaller, more aerobically fit, and run further and faster in practice than running backs, linebackers (LB), and tight ends who form the midskill group.[1,5] In contrast, midskill players tend to perform more blocking, tackling, and accelerations over shorter distances. Thus, various indicators of workload are required to determine how position-specific playing demands impact HRV profiles. Moreover, intense and violent playing demands expose players to sprains, strains, contusions, and other forms of musculoskeletal trauma. In less serious cases, players are cleared by sports medicine staff to maintain participation in a self-limited capacity. However, minor injuries may affect on-field performance outputs or cardiac-autonomic activity[10] thereby potentially confounding associations between HRV and workloads. Therefore, we aimed to quantify position-specific associations between HRV profile characteristics and various external workload parameters in collegiate football players during preseason camp while accounting for injury-related participation status.

Methods

Study Design

We conducted an observational study to evaluate position-specific associations between profiles of movement and cardiac-autonomic activity from a preseason training camp. This is a secondary analysis that includes predominantly unreported data from a previous investigation.[11] Movement and HRV profiles included the intraindividual camp mean and CV for various external training load and cardiac-autonomic parameters, respectively. Injury-related participation status, categorized as "full go" or "go as can," was analyzed and accounted for.

Subjects

Twenty-seven Division 1 players (≥18 y of age) from the defending national championship winning team of the NCAA were included in the analysis. The skill position group (n = 15, height = 187.5 [4.3] cm, weight = 89.8 [5.3] kg) included 8 defensive backs and 7 receivers. The midskill position group (n = 12, height = 188.9 [5.7] cm, weight = 104.6 [6.8] kg) included 5 LBs, 4 running backs, and 3 tight ends. Linemen were excluded due to limited allocation of wearable movement trackers to this position group. Participation in all preseason training sessions, along with training load and HRV data from ≥80% of training days, were considered criteria for inclusion. All players provided written informed consent prior to data collection. Ethical approval for this study was obtained from the University of Alabama institutional review board (Protocol #6253).

Training Camp

Preseason camp involved 15 football training sessions (12 outdoor on natural turf and 3 indoor on artificial turf) over 13 days. Training sessions were 2 to 2.5 hours in duration. Weather conditions were hot and humid during outdoor sessions (heat index: 38.1 [6.5] °C). Protective equipment was progressively phased-in according to heat acclimation guidelines. Further details have been previously described.[11]

Participation Status

Individual participation status, determined by the head athletic trainer, was categorized as unlimited (full go) or self-limited (go as can). The "go as can" category, included players with preexisting conditions (eg, lingering effects of previous surgery or injury), or new issues arising during camp (eg, sprains, strains, and contusions) identified through direct observation by sports medicine staff or player self-reports. These players were cleared to participate within their self-determined or medically advised physical tolerances. Sports medicine staff communicated daily injury status updates to performance staff for documentation on workload reports, which were retrieved from electronic records for analysis.

Cardiac-Autonomic Profiles

Methodology for RHR and HRV acquisition has been previously described.[11] In brief, values were derived from 55-second recordings using a finger pulse sensor (ie, photoplethysmography) and accompanying mobile application (ithlete; HRV Fit LTD). Assessments were preceded by at least 60 seconds for stabilization and were performed while players were seated comfortably on an athletic training table prior to having their ankles taped by athletic training staff. The natural logarithm of the root mean square of successive differences (LnRMSSD) was automatically calculated by the application after automated filtering and was recorded for analysis. The root mean square of successive differences (RMSSD) is a time-domain HRV index that reflects cardiac-parasympathetic modulation and is recommended for use in applied settings with athletes.[12] The intraindividual mean and CV (calculated as [SD/mean] × 100) for RHR and LnRMSSD were calculated to profile intraindividual cardiac-autonomic activity.[8]

Movement Profiles

Physical demands from training sessions were obtained from wearable movement trackers (Catapult Innovations) as previously reported.[11] GPS outputs recorded for analysis included total, high speed (19.4–25.8 $km \cdot h^{-1}$), and sprint (>25.8 $km \cdot h^{-1}$) distances and peak running velocity.[13] Accelerometer outputs included Player Load and total inertial movement analysis (number of accelerations or decelerations >3.5 $m \cdot s^{-2}$).[14] Accelerometer metrics better reflect nonrunning activities such as collisions, changes of direction, and movements over short distances.[14] Player Load has high collinearity with GPS-derived total distance. However, each were included since GPS parameters from indoor sessions are invalid, and practitioners may benefit from seeing values from both metrics. The intraindividual mean and CV for each parameter were calculated to profile workload characteristics.

Statistical Analysis

Normality assumptions were evaluated with Shapiro–Wilk tests. Independent *t* tests or Mann–Whitney U tests were used to compare outcome variables between players grouped by position and participation status. Standardized differences between normally distributed variables were evaluated with Hedges' *g* effect size (ES).[15] Standardized difference values <0.20, <0.6, <1.2, and <2.0 were considered *trivial*, *small*, *moderate*, and *large*, respectively.[16] Pearson correlations were used to quantify bivariate associations between movement and HRV profile metrics. Correlation coefficients <.10, <.3, <.5, <.7, <.9, and >.9 were considered *trivial*, *small*, *moderate*, *large*, *very large*, and *near perfect*, respectively.[16]

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If significant opposing associations were observed between position groups for the same metric, multiple regression was used to investigate the possibility of an interaction effect with participation status included as a covariate. Data are reported as mean (SD) or median (interquartile range). $P < .05$ were considered statistically significant. Analyses were conducted with JMP (version 16, SAS Institute Inc).

Results

Total distance, high-speed distance, sprint distance, peak velocity, Player Load, and RHR mean differed between position groups ($P < .05$), whereas no differences were observed for CV variables ($P > .05$). Between-position group values and comparison statistics are presented in Table 1.

Injury type, body location, and number of days listed as "go as can" are presented in Table 2. No between-participation status group differences ($P > .05$) were observed for any outcome variables (Table 3), including body mass (95.9 [9.6] vs 97.1 [9.9] kg; $P = .74$). However, ES analysis showed that LnRMSSD was *moderately* greater among players with full go status ($P < .07$).

In skill players, LnRMSSD CV was inversely associated with total distance and Player Load and positively associated with total distance CV ($P < .05$). Additionally, LnRMSSD was inversely associated with total distance CV ($P < .05$). In midskill players, LnRMSSD CV was positively associated with total distance, Player Load, high speed CV, and total inertial movement analysis CV and inversely associated with high-speed distance and sprint distance ($P < .05$). Additionally, RHR CV was inversely associated with high-speed distance and sprint CV, while RHR was positively associated with Player Load. Correlation coefficients are presented in Table 4.

Table 1 Variable Mean (SD) or Median (Interquartile Range) and Comparison Statistics Between Position Groups

Variable	Skill	Midskill	*P*	ES	Magnitude
Total distance, m	5712 (636)	4902 (357)	**<.001**	1.48	*Large*
High-speed distance, m	316 (75)	171 (81)	**<.0001**	1.81	*Large*
Sprint distance, m	95 (38)	38 (23)	**<.0001**	1.71	*Large*
Peak velocity, km·h^{-1}	30.1 (1.4)	28.1 (1.4)	**<.001**	1.39	*Large*
PlayerLoad, au	567 (80)	508 (35)	**.020**	0.89	*Moderate*
Total IMA, au	50 (9)	57 (10)	.073	–0.72	*Moderate*
LnRMSSD mean	4.1 (0.4)	3.9 (0.4)	.279	0.48	*Small*
RHR mean	66.0 (6.2)	71.9 (7.2)	**.034**	–0.86	*Moderate*
Total distance CV, %	11 (10)	10 (8)	.826	—	—
High speed CV, %	34 (12)	45 (21)	.116	–0.64	*Moderate*
Sprint CV, %	54 (19)	67 (26)	.162	–0.56	*Small*
Peak velocity CV, %	7 (3)	6 (2)	.842	0.37	*Small*
PlayerLoad CV, %	14 (8)	12 (4)	.272	—	—
Total IMA, CV%	28 (8)	25 (7)	.337	0.38	*Small*
LnRMSSD CV, %	7.3 (2.1)	8.2 (4.1)	.496	–0.28	*Small*
RHR CV, %	8.9 (2.2)	8.8 (3.6)	.891	0.03	*Trivial*

Abbreviations: au, arbitrary units; CV, coefficient of variation; ES, effect size; IMA, inertial movement analysis; LnRMSSD, natural log of the root mean square of successive differences; RHR, resting heart rate. Bold *P* values are statistically significant (<.05).

Table 2 Description of Individual Injury Type, Day of Incident, and Number of Impacted Days

Player	Group	Injury	Day of incident	"Go as can" days
A	Skill	Shoulder stinger	8	1
B	Skill	Hamstring strain	8	1
C	Skill	Previous tibia fracture	1	9
D	Skill	Shoulder stinger	6	2
E	Skill	Hamstring soreness	3	1
F	Skill	Foot soreness	1	1
G	Midskill	Groin soreness, knee sprain	3, 6	2
H	Midskill	Unlisted	7	1
I	Midskill	Groin strain	8	1
J	Midskill	Unlisted	6	1
K	Midskill	Partial quadriceps tear	5	5
L	Midskill	Unlisted	8	2

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Table 3 Variable Mean (SD) or Median (Interquartile Range) and Comparison Statistics Between Participation Status Groups

Variable	Full go	Go as can	*P*	ES	Magnitude
Total distance, m	5500 (700)	5167 (591)	.192	0.49	*Small*
High-speed distance, m	250 (113)	250 (102)	.962	0.00	*Trivial*
Sprint distance, m	69 (46)	70 (42)	.908	–0.02	*Trivial*
Peak velocity, km·h^{-1}	29.2 (1.7)	29.3 (1.6)	.927	–0.06	*Trivial*
PlayerLoad, au	551 (77)	529 (60)	.415	0.30	*Small*
Total IMA, au	52 (9)	54 (12)	.688	–0.19	*Trivial*
LnRMSSD mean	4.2 (0.5)	3.9 (0.2)	.069	0.73	*Moderate*
RHR mean	66.9 (7.3)	70.7 (6.7)	.175	–0.52	*Small*
Total distance CV, %	9.6 (7.7)	14.0 (10.0)	.450	—	—
High speed CV, %	35.6 (25.7)	35.9 (21.6)	.317	—	—
Sprint CV, %	56.7 (26.4)	62.8 (18.4)	.480	–0.25	*Small*
Peak velocity CV, %	6.0 (2.5)	5.4 (4.2)	.788	—	—
PlayerLoad CV, %	12.5 (4.0)	14.5 (5.4)	.309	–0.42	*Small*
Total IMA, CV%	26.2 (7.3)	27.8 (8.2)	.609	–0.20	*Small*
LnRMSSD CV, %	6.3 (3.5)	7.4 (4.8)	.272	—	—
RHR CV, %	9.1 (3.0)	8.7 (2.8)	.726	0.13	*Trivial*

Abbreviations: au, arbitrary units; CV, coefficient of variation; ES, effect size; IMA, inertial movement analysis; LnRMSSD, natural log of the root mean square of successive differences; RHR, resting heart rate.

Table 4 Correlation Coefficients Between Resting-Heart-Rate Parameters and Movement Variables by Position Group

	RHR		RHR CV		LnRMSSD		LnRMSSD CV	
Variable	Skill	Midskill	Skill	Midskill	Skill	Midskill	Skill	Midskill
Total distance	–.45	.23	–.47	.32	.47	–.06	–.67*	.60*
High speed	.26	–.35	–.38	–.71*	–.29	–.04	.22	–.74*
Sprint	.43	–.24	–.20	–.46	–.45	.09	.30	–.61*
Peak velocity	.09	–.27	–.32	–.35	–.14	.05	–.18	–.50
Player Load	–.39	.61*	–.39	.37	.49	–.47	–.56*	.63*
Total IMA	–.19	.15	.06	.13	.22	–.21	–.17	–.01
Total distance CV	.33	.33	.03	–.20	–.52*	.14	.61*	–.03
High speed CV	.13	.36	.07	.51	–.34	–.28	.39	.68*
Sprint CV	–.42	.31	.15	–.63*	.37	–.33	–.03	–.36
Peak velocity CV	.11	.22	.17	.01	–.21	.06	.35	.07
Player Load CV	.41	.34	–.14	.07	–.47	.23	.57*	.26
Total IMA CV	–.13	.53	.08	.27	.08	–.07	.22	.66*

Abbreviations: CV, coefficient of variation; LnRMSSD, natural log of the root mean square of successive differences; RHR, resting heart rate.
*Statistically significant association ($P < .05$).

Multiple regression analyses, controlling for injury-related participation status, showed significant position × total distance (model effect $P = .018$, adjusted $r^2 = .30$, interaction $P = .002$) and position × Player Load (model effect $P = .020$, adjusted $r^2 = .29$, interaction $P = .002$) interactions for LnRMSSD CV, such that as each load metric increased; LnRMSSD CV decreased among skill players but increased among midskill players (Figure 1). For both models, residuals followed a normal distribution ($P = .254$–$.494$), and the assumption of homoscedasticity was confirmed.

Discussion

This study evaluated associations between profiles of external workload and cardiac-autonomic activity throughout preseason college football training while accounting for injury-related participation status. The main finding was that associations between markers of total workload and LnRMSSD CV varied as a function of position group. Skill players who ran the greatest total distances and accumulated the greatest Player Loads tended to display the most stable HRV (ie, lower LnRMSSD CV). Conversely, midskill players who ran the greatest total distances and accumulated the greatest Player Loads displayed the least stable HRV. These contrasting associations were independent of injury-related participation status. However, consistent between positional groups, players with greater daily variation in select workload parameters also showed greater daily variation in LnRMSSD. Furthermore, minor injuries did not systematically affect workload outputs, but a qualitatively *moderate* negative

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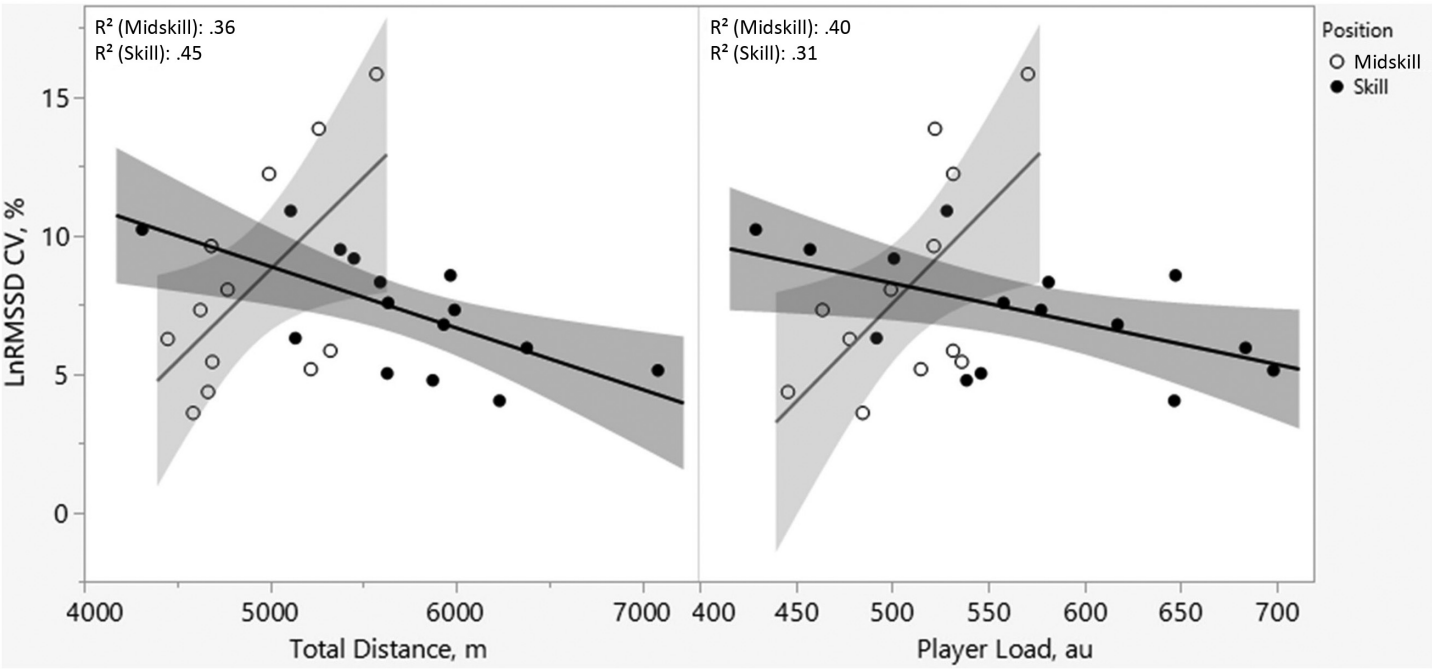

Figure 1 — Scatterplots showing position group × total distance and position group × PlayerLoad interactions. LnRMSSD CV = coefficient of variation of the natural log of the root mean square of successive differences.

effect on LnRMSSD was observed for players with "go as can" participation status.

Skill players with a lower LnRMSSD CV performed consistently high total movement demands, reflected in *large* inverse associations with total distance and Player Load, and *large* positive associations with their CV values. A similar association between LnRMSSD CV and Player Load ($r = -.60$, $P < .05$) involving players from all position groups was observed during spring camp, when the weather was cooler and training was less frequent.[6] The strongest known performance correlate of low LnRMSSD CV in athletes is high aerobic fitness,[17,18] which may help explain our findings in skill players. Although we did not quantify aerobic fitness, higher total distances in training are associated with greater aerobic fitness in team-sport athletes.[19] Moreover, faster heat acclimation (ie, days to plateaued core temperature),[20] lower peak core temperatures,[21] and lower cardiovascular strain during exercise heat stress are observed in individuals with higher aerobic fitness, including football players.[22] This may be because fitter athletes exhibit characteristics of heat acclimated individuals and are considered partially heat acclimated.[23] Accordingly, expected physiological adaptations that affect LnRMSSD may have already occurred in players with higher fitness, or occurred faster during camp, contributing to a more stable LnRMSSD profile. Reduced LnRMSSD CV with increasing workloads have previously been associated with increased aerobic fitness[8] and a reduced internal-to-external training load ratio in team-sport athletes.[17] Moreover, higher aerobic fitness is associated with accelerated posttraining cardiac-parasympathetic reactivation,[12] which may contribute to less training-induced day-to-day fluctuation in LnRMSSD. Thus, we speculate that an ability to perform consistently high training loads during preseason camp in the heat while maintaining stable LnRMSSD may reflect a combination of high initial aerobic fitness and positive cardiovascular and autonomic adaptation to training.

Unlike associations observed in skill players, midskill players with higher LnRMSSD CV performed higher total workloads, reflected in *large* positive associations with mean total distance and Player Load. Midskill players are larger, slower, and less aerobically fit than skill players.[5] Consequently, midskill players experience greater sweat rates (1.98 [0.48] vs 1.40 [0.45] $L \cdot h^{-1}$) and fluid losses (8.9 [2.2] vs 6.4 [2.0] L) than skill players[24] and are at greater risk of exertional heat illness.[2] Thus, higher body mass[6] and lower aerobic fitness,[12] combined with greater heat stress during training may promote physiological responses that affect cardiac-autonomic regulation to a greater extent in larger players exposed to higher training loads. For instance, daily pretraining assessment of plasma volume has been shown to vary between approximately −5% to +9% in college football players through preseason camp.[25] Moreover, dehydration-induced hypovolemia and exercise heat stress reduce RMSSD,[26] whereas subsequent training-induced plasma volume expansion correlates with increased RMSSD.[27] Thus, greater daily reductions and subsequent rebounding in LnRMSSD may help explain higher CV values among heavily loaded midskill players. Meanwhile, midskill players with higher LnRMSSD CV also tended to perform lower high-speed running and sprint distances. Notably, higher total distance was associated with lower high-speed running, sprint distance, and peak velocity ($r = -.57$ to $-.77$; $P < .05$) in midskill but not skill players ($P > .70$). Higher total workloads may stem from additional special teams practice for selected players (eg, kickoff or return teams) beyond position-specific drills, potentially contributing to fatigue-related reductions in high-speed output, discussed further below.

Irrespective of position group, players with greater daily variation in LnRMSSD showed greater daily variation in various movement parameters. Specifically, skill players with higher LnRMSSD CV tended to show higher CV values for total workload, whereas midskill players with higher LnRMSSD CV tended

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to show higher CV values for high-intensity workload. Notably, for midskill, 2 middle LBs displayed the highest LnRMSSD CV on the team (13.8% and 15.8%) and performed Player Load values (522 and 571 au) comparable with or greater than the mean of skill players (567 au), despite having substantially greater body mass (~110 kg vs 89 kg). Moreover, Player Load values from these LBs were substantially greater than mean preseason Player Load values of LBs from a previous study (419 au) of a comparable college football team,[13] suggesting very high workloads for these players. Midskill playing demands involve a higher volume of severe impact collisions relative to skill,[28] potentially exposing players with higher workloads to more damage-associated inflammation and subconcussive impacts, which can affect cardiovascular[29] and autonomic function.[30] Additionally, increases in LnRMSSD CV as a result of increased training load have been associated with decrements in running performance[8] and subjective recovery markers.[31] Consistent with an interpretation of fatigue, midskill players with higher LnRMSSD CV showed less day-to-day consistency in high-speed running and total inertial movement analysis. Similarly, daily variation in perceived fatigue and LnRMSSD have been associated with daily variation in high-intensity running distance in an elite men's soccer team, linking fluctuations in HRV with fatigue-related fluctuation in on-field performance output.[32] However, associations between LnRMSSD CV and total workload parameters were not perfect. For example, 2 tight ends performed total workloads comparable to the aforementioned LBs yet maintained more stable LnRMSSD (CV ≤ ~6%, Figure 1). Thus, even players of similar size and total workload can exhibit interindividual variation in LnRMSSD responses; but, whether variation in LnRMSSD or intense movement was an effect of daily changes in training contents, coaching decisions (ie, repetition distribution among players), fatigue, fitness, or other factors is unclear.

No significant differences in workload or cardiac-autonomic parameters were observed between players categorized as "full go" and "go as can." Moreover, ES magnitudes were *trivial* to *small*, with one exception: LnRMSSD for "full go" was *moderately* higher than "go as can" ($P < .07$). Previous research examining associations between HRV and nonconcussion-related athletic injuries are limited. One investigation reported a greater risk of overuse problems when weekly averaged LnRMSSD was reduced in response to increased training load (relative risk: 2.61; 90% confidence interval, 1.38–4.93) in a small group of competitive CrossFitters (n = 6).[33] Another small investigation in nonathletes reported lower HRV and higher sympathetic activation in patients experiencing mild to moderate pain within 1 week of injury relative to a control group.[34] These findings[33,34] may support the hypothesis that musculoskeletal injury can be reflected in reduced HRV via altered autonomic activation to accommodate needs for increased blood supply and inflammatory neuromediators at the site of damaged tissue.[10] However, whether reduced HRV precedes injury has not been well explored. One investigation reported progressive reductions in RMSSD (time effect: $P < .05$) and a greater incidence of injuries per 1000 hours of training ($P < .05$) in adolescent sprint swimmers during a macrocycle relative to distance swimmers who maintained stable RMSSD.[35] In a small case comparison of linemen from the current football team, acute soft tissue injuries were observed in players with suppressed LnRMSSD during the competitive period.[7] Thus, in addition to potential effects of injury on cardiac-autonomic activity,[10] reduced LnRMSSD may also reflect a heightened risk of injury resulting from accumulating stress and fatigue. Accordingly, we hypothesize that the association between HRV and injury may be bidirectional.

The current study has several limitations. Injury status was determined subjectively by sports medicine staff, and players may not consistently self-report medical issues. Additionally, collection of HRV data at the training facility increases noise potential versus postwaking conditions. Inclusion of a standardized neuromuscular test would have helped determine if overreaching explained any of the associations between workload and HRV profiles, although such tests may also be confounded by minor injuries. Peak running velocity, or its consistency across days, may be considered a performance indicator in certain contexts, but neither mean nor CV values were significantly associated with HRV parameters. Additionally, explanations for HRV responses being potentially related to factors such as aerobic fitness and plasma volume changes are speculative based on previous research as we were unable to collect these markers from players. Lack of internal load assessment and sample size are further limitations that should be addressed in future research. Finally, it was beyond our aim to further interrogate the potential association between injury status and HRV, given that players with more severe injuries were excluded due to limited training participation. However, our observation that "go as can" players had *moderately* lower LnRMSSD than "full go" players ($P < .07$) warrants future investigation. Specifically, we need a further understanding of how various injuries affect HRV, how HRV evolves throughout recovery, and how changes in HRV (eg, fatigue-related decrements) may precede and impact injury risk. These insights would help further elucidate the potential role of HRV tracking in supporting player health and performance.

Practical Applications

When interpreting cardiac-autonomic responses to preseason training, practitioners should consider whether LnRMSSD values are becoming more or less stable in response to high external workloads. The CV is an easily calculated metric of daily variation that demonstrated position-specific associations with external workload parameters and can be used to compare interindividual responses. Within the skill position group, players who performed the highest total workloads also tended to exhibit less daily variation in this output along with more stable HRV. Thus, stable LnRMSSD in this context seems to reflect a high and/or rapidly improving capacity for physical training in the heat with minimal disturbance in cardiac-parasympathetic activity. Within the midskill position group, players who performed the highest total workloads tended to exhibit greater daily variation in high-intensity movement and LnRMSSD. A potentially important implication of this finding is that midskill players may respond better (ie, display more stable HRV and maintain high-intensity output) with more moderate total workloads. Accordingly, close monitoring of heavily loaded midskill players exhibiting substantial fluctuation in LnRMSSD seems advisable, though further study is needed to support these interpretations. While LnRMSSD CV was more consistently associated with workload parameters, a potentially negative influence (ie, *moderate* ES, $P < .07$) of "go as can" status was observed for mean LnRMSSD. Thus, injury-related participation status should be considered when examining and interpreting interindividual differences in average LnRMSSD.

Conclusions

This study demonstrated that associations between heart-rate-variability (HRV) profiles and workload parameters vary between

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position groups of a college football team during preseason training camp. Skill players who consistently performed higher total workloads tended to exhibit more stable daily HRV. In contrast, midskill players who performed the greatest total workloads tended to show greater daily fluctuation in HRV and high-intensity movement, though some interindividual variation was observed. These position-specific associations between HRV and workload profiles were not confounded by injury-related participation status. However, a *moderate* effect-size difference in LnRMSSD between players categorized as "full go" and "go as can" suggests that minor injuries may have a nontrivial negative influence on cardiac-parasympathetic activity, or vice versa. Collectively, findings suggest that HRV profiles may be useful in evaluating training responses during preseason camp, potentially supporting player health and performance through physiologically informed workload-management strategies.

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